A Space CoBot for personal assistance in space stations*

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Abstract

Microgravity environments inside inhabited space stations are particularly challenging for service robots, being quite different from Earth in various ways. In this paper we propose and discuss the design and applicability of a collaborative aerial robot, called Space CoBot, to provide personal assistance to astronauts in microgravity environments. The platform is based on a modular holonomic aerial vehicle with an hexarotor configuration. In this paper we explore the feasibility of using Space CoBot for performing two tasks: (1) scavenging afloat waste and (2) stabilization of astronaut motion. Scavenging is performed by tracking and sucking debits, while stabilization aims at driving to zero the motion of an astronaut attached to the robot. A convergent motion controller is used for guidance of the vehicle to accomplish both tasks. We provide simulation results on a realistic simulator to illustrate the feasibility of the approach.

1 Introduction

Transporting to and maintaining astronauts in orbiting space stations is hard: space flight is risky and the costs associated with life maintenance are extremely high. Thus, it is desirable that crew sizes are kept small. In this context, the use of autonomous service robots that perform collaborative tasks with humans (CoBots) is particularly interesting.

In this paper we explore the design and the application of a robot, called Space CoBot, for collaborative tasks with astronauts in the interior of a space station. These are microgravity environments with breathable air at atmospheric pressure, thus allowing propeller based propulsion. Space CoBot exploits these properties being an aerial vehicle based on a hexarotor design. This design allows for a fully omnidirectional (holonomic) motion (6 Degrees of Freedom - 6-DoF), while being based in electrical motors and propellers is mechanically simple and reliable.

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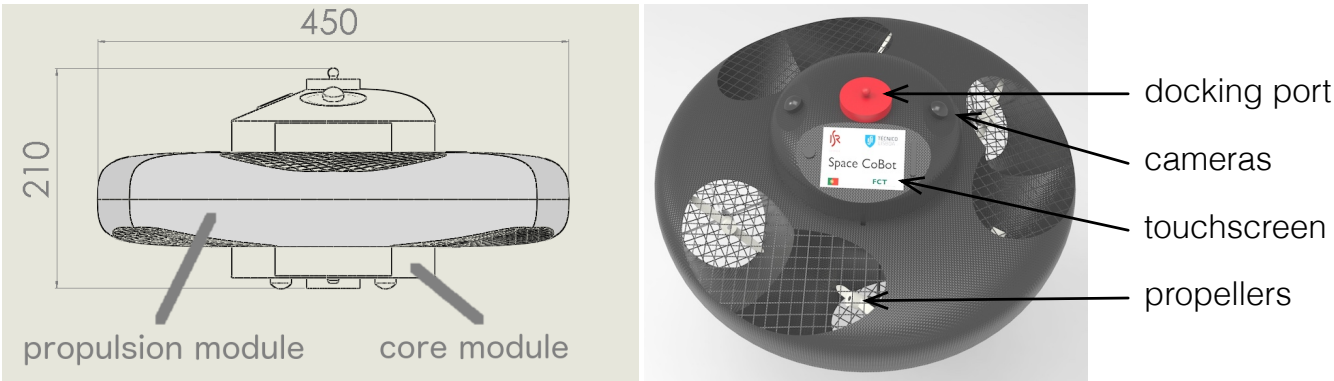
The idea of aerial vehicles inside space stations is not new. The NASA project SPHERES (Synchronized Position Hold Engage Reorient Experimental Satellites) started in 2000 with the design of a small pressurized air propulsion vehicle [Miller *et al.*, 2000]. In 2006 three SPHERES units were deployed aboard the International Space Station (ISS) [Nolet *et al.*, 2007]. More recently, NASA proposed the Astrobee vehicle, with a simpler propulsion system based on several centrifugal fans and nozzles [Bualat *et al.*, 2015]. The Astrobee also features a 2-DoF arm and docking capability.

This paper is structured as follows: sections 2 and 3 describe the design of Space CoBot and the proposed motion controller, summarizing [Roque and Ventura, 2016], sections 4 and 5 describe our approach to the debris scavenging and astronaut stabilization tasks, simulation results are presented in section 6, and section 7 wraps up the paper with some conclusions and future work.

2 Vehicle Design

The design of Space CoBot is modular, comprising an outer propulsion module and an inner core module, illustrated in Figure 1. The propulsion module comprises six propellers, arranged in such a way the motion is fully holonomic. We further detail this module below. The core module contains the power supply (battery), interface electronics, computers, a touchscreen, an array of video cameras for perception and telepresence, and an inertial measurement unit (IMU). This module can be easily extended with other components, such as a robotic arm. We consider vision-based navigation, using the camera array. At the opposite sides of the core module we include two docking ports, for two purposes: (1) charging the batteries and (2) attachment to another Space CoBot (see Figure 2). The design of these docking ports follows previous work described in [Marques *et al.*, 2007].

The propulsion module is based on a hexarotor design, where the rotation axes of the six propellers are not parallel. Rather, each propeller $i = 1, \ldots, 6$ has an angular offset ϕ_i from the Z axis, as shown in Figure 3. Each such propeller contributes with a force $\bar{F}_i$ and a torque $\bar{M}_i$ on the robot center of mass (CoM), given by the following expression [Roque



(a) (b)

Figure 1: Design of the Space CoBot: (a) side view with the core and propulsion modules indicated with dimensions in millimeters; (b) CAD rendering showing some of the main components.

Figure 2: Render of two Space CoBots rigidly attached through their docking ports.

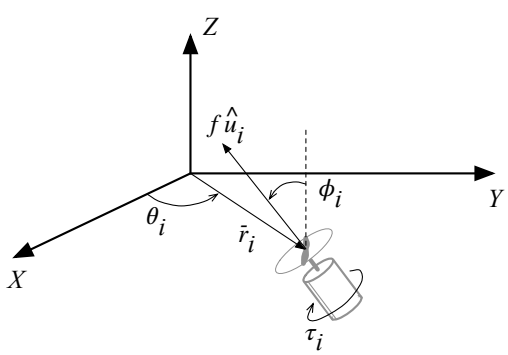


Figure 3: Notation used for modeling a single propeller with respect to the body frame of the robot, centered on its center of mass.

and Ventura, 2016], following the notation of this figure:

$$\begin{pmatrix} \bar{F}_i \\ \bar{M}_i \end{pmatrix} = \bar{a}_i\, u_i \qquad \bar{a}_i = \begin{pmatrix} K_1 \hat{u}_i \\ K_1 \bar{r}_i \times \hat{u}_i - w_i K_2 \hat{u}_i \end{pmatrix} \tag{1}$$

where $\times$ denotes vector cross product, u_i is the actuation signal, and w_i is a flag with value either -1 or 1 depending on whether the propeller rotates clockwise or anti-clockwise for a positive thrust long $\hat{u}_i$. Constants K_1 and K_2 are given by

$$K_1 = \rho D^4 C_T \qquad K_2 = \frac{\rho D^5}{2\pi} C_P \tag{2}$$

where ρ is the air density, D is the propeller diameter, and the C_T and C_P are blade dependent adimensional constants called thrust and power coefficients, following the momentum-blade element theory [McCormick, 1995]. The actuation signal is $u_i = n_i^2$, where n_i is the blade revolutions per second of propeller i. We will adopt u_i as the input signal for the i-th motor controller, to maintain a linear relation between actuation and forces/torques.

Since the contributions of each propeller is expressed in the same reference frame, the net force and torque will be given by the sum of these contributions. This sum can be put in matrix form as

$$\begin{pmatrix} \bar{F} \\ \bar{M} \end{pmatrix} = \mathbf{A}\, \bar{u} \tag{3}$$

where $\mathbf{A} = [\bar{a}_1 \cdots \bar{a}_N]$ is a square matrix, hereby called *actuation matrix*, and $\bar{u} = [u_1 \cdots u_N]^T$ is the actuation input vector. The crucial observation is that, if the actuation matrix $\mathbf{A}$ is at least rank 6, the linear equation (3) can be solved for $\bar{u}$ for any given combination of $\bar{F}$ and $\bar{M}$. A necessary (but not sufficient[1]) condition for this to be true is to have at least 6 propellers, thus justifying the hexarotor design. Note that this matrix only depends on these parameters: angles $\{\theta_i\}$ and $\{\phi_i\}$, distance d, the trust coefficients K_1, K_2, and $\{w_i\}$.

[1]Sufficiency requires $\mathbf{A}$ to be full rank.

As mentioned before, if the axes of all propellers were parallel, $\{\phi_i = 0\}$, holonomy would be lost. The question is then which angle values to choose. Considering that the thrust is bounded, the value of these angles hold a direct relationship with the maximum force and torque attainable along an arbitrary direction. For instance, when all propeller axes are parallel, one gets maximum thrust along the Z axis, but zero along any orthogonal direction; as these angles depart from zero, we are able to tradeoff the maximum thrust along Z with non-zero maximum thrust along any given orthogonal directions. In [Roque and Ventura, 2016] we addressed this problem by formulating it as a multi-criteria optimization problem. We summarize the main results below.

We consider each actuation signal to be bounded between -1 and 1, that is,

$$-1 \leq u_i \leq 1 \quad \text{for} \quad i = 1, \ldots, 6 \tag{4}$$

since (3) is linear, and thus the optimization problem is invariant to the scaling of these signals. According to (3), this hypercube will map onto a 6-dimensional convex polyhedron[2] in the $(\bar{F}, \bar{M})$ space. Our goal will be to find the configurations of angles $\{\phi_i\}$ and flags $\{w_i\}$ that maximize the range of forces (and torques) over all directions. Geometrically, this corresponds to changing $\{\phi_i\}$ and $\{w_i\}$ such that a ball of nonzero radius can fit inside the 3-dimensional convex polyhedron in the $\bar{F}$ space mapped by the actuation hypercube in (4), while keeping zero torque, $\bar{M} = 0$. A similar reasoning applies to the torque space $\bar{M}$, while keeping $\bar{F} = 0$.

Since we intend to both maximize force and torque, we make the trade-off between the two explicit by taking a multi-criteria optimization approach. The formulation of this problem is detailed in [Roque and Ventura, 2016], having the following form:

$$\begin{aligned} &\text{minimize } (p, q) \\ &\text{subject to:} \\ &\quad p \geq \|b_i\|^2, i = 1, \ldots, 6 \\ &\quad q \geq \|c_i\|^2, i = 1, \ldots, 6 \end{aligned} \tag{5}$$

In this problem, the optimization variables are $\{p, q\} \cup \{\phi_i\} \cup \{w_i\}$ and the cost functions are p and q. For the sake of symmetry we kept the angles $\{\theta_i\}$ equally spaced in 60 deg intervals. Intuitively, p^{-1} and q^{-1} correspond to the radius of the above mentioned balls in the $\bar{F}$ and $\bar{M}$ spaces, respectively. Minimizing p or q, subject to the problem constraints, correspond to maximizing the radius of the balls that fit inside the polyhedrons mapped by the bounded actuation.

The solution of this multi-criteria optimization problem is the set $\mathcal{P}$ of non-dominated solutions, also known as the *Pareto optimal set* [Statnikov and Matusov, 1995]. We obtained a numerical approximation to this set, from which we chose a solution maximizing the force component, since the corresponding maximum torque is not significantly lower than other non-dominated solutions [Roque and Ventura, 2016]. This solution[3] is shown in Table 1. All of the following results shown in this paper use this selected configuration.

propeller (i)	1	2	3	4	5	6
θ_i	0	60	120	180	240	300
ϕ_i	55	-55	55	-55	55	-55
w_i	-1	1	-1	1	-1	1

Table 1: Design parameters of the selected solution. Both $\{\theta_i\}$ and $\{\phi_i\}$ are expressed in degrees.

3 Position and Attitude Control

The dynamical model of the vehicle can be derived from the Newton and Euler equations of motion. Let us denote the position and velocity of the body frame $\mathcal{B}$, centered on the vehicle CoM and aligned according to Figure 3, with respect to the inertial frame $\mathcal{I}$ as $\bar{x}$ and $\bar{v}$, the rotation matrix of frame $\mathcal{B}$ with respect to $\mathcal{I}$ as $\mathbf{R}$, and the angular velocity of the vehicle in the body frame $\mathcal{B}$ as $\bar{\omega}$. Then,

$$\begin{cases} \dot{\bar{x}} = \bar{v} \\ m\dot{\bar{v}} = \mathbf{R}\bar{F} \\ \dot{\mathbf{R}} = \mathbf{R}\,\mathbf{S}(\bar{\omega}) \\ \mathbf{J}\dot{\bar{\omega}} = \bar{M} - \bar{\omega} \times \mathbf{J}\bar{\omega} \end{cases} \tag{6}$$

where the constants m and $\mathbf{J}$ are the vehicle's mass and moment of inertia, while $\mathbf{S}(\bar{\omega})$ is the skew-symmetric matrix defined by

$$\mathbf{S}(\bar{\omega}) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \tag{7}$$

for $\bar{\omega} = [\omega_x\, \omega_y\, \omega_z]^T$. Note the absence of the gravity force in this model.

The approach used for the motion control of the vehicle exploits its holonomic design by decoupling the translational and rotational modes. To do so, we first apply feedback linearization [Sastry, 1999] to the translational part of (6):

$$\begin{cases} \dot{\bar{x}} = \bar{v} \\ \dot{\bar{v}} = \bar{p} \\ \bar{F} = m\mathbf{R}^T\bar{p} \end{cases} \tag{8}$$

and then design a feedback controller for $\bar{p}$. Since this dynamical system is diagonal and second order, a PD controller is enough to ensure exponential convergence:

$$\begin{aligned} \bar{e}_x &= \bar{x} - \bar{x}_d \\ \bar{e}_v &= \bar{v} - \bar{v}_d \\ \bar{p} &= -k_x\bar{e}_x - k_v\bar{e}_v \end{aligned} \tag{9}$$

where $\bar{x}_d$ and $\bar{v}_d$ are the desired position and velocity vectors in the inertial frame $\mathcal{I}$, and k_x and k_v are the proportional and derivative gains of the PD controller.

[2] A convex polyhedron is an intersection of a finite number of half-spaces.

[3] The angles are rounded off to the nearest degree unit for the sake of simplicity. However, the impact of this on the cost is minimal.

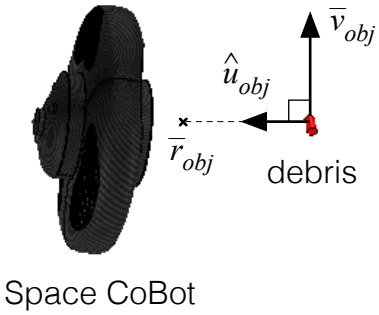


Figure 4: Geometry of the debris scavenging problem.

For the attitude control we follow the exponentially convergent $SO(3)$ controller proposed in [Lee, 2012]:

$$\begin{aligned}
\bar{e}_R &= \frac{1}{2\sqrt{1+tr[\mathbf{R}_d^T\mathbf{R}]}} S^{-1}(\mathbf{R}_d^T\mathbf{R} - \mathbf{R}^T\mathbf{R}_d) \\
\bar{e}_\omega &= \bar{\omega} - \mathbf{R}^T\mathbf{R}_d\,\bar{\omega}_d \\
\bar{M} &= -k_R\,\bar{e}_R - k_\omega\,\bar{e}_\omega \\
&\quad + S^{-1}(\mathbf{R}^T\mathbf{R}_d\bar{\omega}_d)\mathbf{J}\mathbf{R}^T\mathbf{R}_d\bar{\omega}_d + \mathbf{J}\mathbf{R}^T\mathbf{R}_d\dot{\bar{\omega}}_d
\end{aligned} \tag{10}$$

where $\mathbf{R}_d$ is the rotation matrix of the desired attitude, $\bar{\omega}_d$ is the desired angular velocity vector, with k_R and k_ω as controller gains. The S^{-1} function performs the inverse operation as the one defined in (7), that is, recovers the vector from a given skew-symmetric matrix.

4 Debris Scavenging

One distinctive feature of microgravity environments is that objects do not fall down to the floor when unreleased from one's hand. In a space station environment, unattached objects, such as food debris, pens, and even liquids, fly freely until coliding with something else, e.g., a wall. Therefore, astronats are required to manage manually such objects, demanding additional cognitive effort. We propose here the automatic management of these objects by Space CoBot. That is, we consider this robot to be able to detect and track free flying debris, e.g., by the use of its camera array, and to be equipped with a small vacuum cleaner capable of capturing and storing them internally.

To provide a proof of concept for this task, we considered a small object flying freely in an uncluttered environment. Since it is free, the linear and angular velocities are constant. To capture it we consider a three phase process: first, the Space CoBot matches its trajectory at a distance D_1 of the object; second, it approaches the object at a constant velocity with respect to it until a threshold distance D_2 is achieved, where we consider it to be within reach of a small onboard vacuum cleaner; finally, the object is sucked into the vehicle by it.

The geometry of the problem is shown in Figure 4. Let $\bar{x}_{obj}$ and $\bar{v}_{obj}$ be the object position and velocity. We define $\hat{u}_{obj}$ to be the unit vector orthogonal to $\bar{v}_{obj}$ which $\bar{x}_{obj}+\hat{u}_{obj}$ is closest to the Space CoBot position. The position reference for tracking the object, $\bar{r}_{obj}$, is defined along $\hat{u}_{obj}$, that is:

$$\bar{r}_{obj} = \bar{x}_{obj} + d\,\hat{u}_{obj} \tag{11}$$

In the first phase of the process, the Space CoBot matches its trajectory by tracking $\bar{x}_d = \bar{r}_{obj}$ and $\bar{v}_d = \bar{v}_{obj}$ with $d = D_1$. In the second, it approaches the object at a constant velocity along $\hat{u}_{obj}$ by decreasing d at a constant rate from D_1 to D_2. Finally, when the object is sufficiently close to the reference at $d = D_2$, it is sucked into the vehicle.

5 Astronaut Stabilization

In microgravity, it is hard to keep the same position through time, as every little touch on indoor surfaces create a linear velocity and angular momentum on the astronauts body. To overcome such problem, there are steel bars throughout the space station, which one can grab and stabilize. Although this is a simple solution, there are situations in which these cannot be used, either because the user is performing a manual operation or because these bars are out of range.

Targeting these situations, we studied the use of a Space CoBot attached to the back of the astronaut, like a backpack, that actively uses its propulsion to reduce the astronaut velocity to zero, thus stopping him from drifting. This is performed by using velocity feedback only, that is, setting $\bar{v}_d = \bar{\omega}_d = 0$ and disabling the position/attitude feedback with $k_x = k_R = 0$. By doing so, the controller will counteract any non-zero motion of the astronaut, both translational and rotational, while ignoring its position.

6 Simulation Results

The realistic simulator V-REP [Rohmer *et al.*, 2013] was used to validate the approach. Results showing convergence of the controller to arbitrary setpoints have been previously shown in [Roque and Ventura, 2016], including sensitivity to localization noise and to unmodeled dynamics, e.g., an heavy load attached to the robot. In this paper we focus on showing simulations of the debris scavenging and astronaut stabilization tasks.

Figure 5 shows a simulation result of the debris scavenging task. The target object is moving at a constant velocity of 0.03m/s along the Y axis, while Space CoBot goes through the three phases described above, also indicated in the figure. The distance between the Space CoBot and the debris is plotted in this figure. Two plateaus are visible: at $D_1 = 0.4m$, where Space CoBot matches the debris velocity, and at $D_2 = 0.1m$, when we consider the object to be reachable by the robot.

To simulate the astronaut stabilization task, we used the Astri humanoid model, bundled with V-REP, to emulate the body dynamics of an astronaut. Its total mass is 73Kg (12 times heavier than Space CoBot). We tested two situations, illustrated in Figure 6: in (a) the astronaut stretches his legs and hits the top of a table, while in (b) the astronaut stretches his arms and hits a vertical wall. In both cases, the hit provokes a reaction momentum impelling the astronaut on the opposite direction to the hit.

For each one of these situations we compared the trajectories of Space CoBot, always attached to the astronaut, with

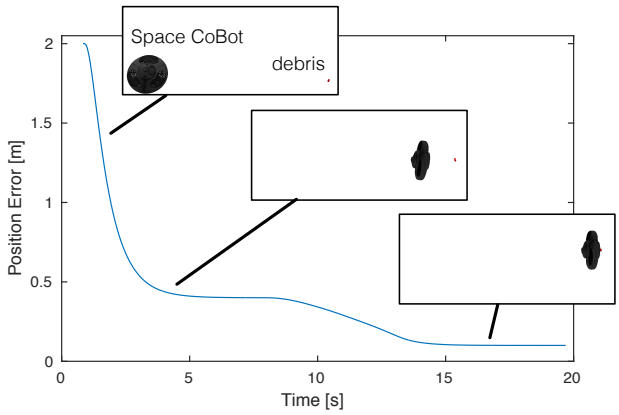
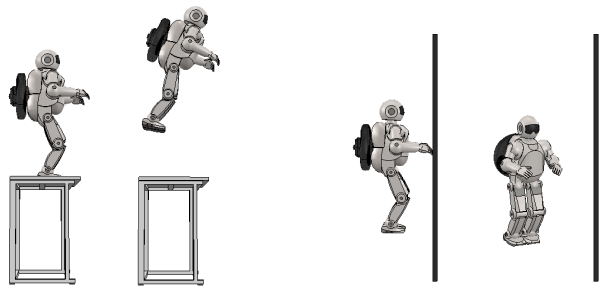
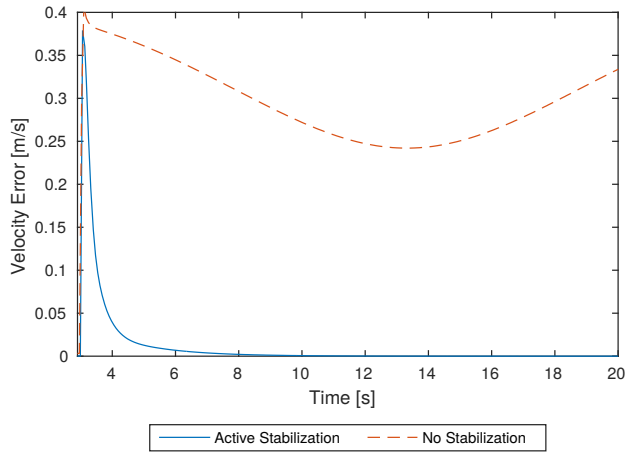


Figure 5: Simulation results for debris scavenging: distance from Space CoBot to the object position, along the three phases explained in the text.

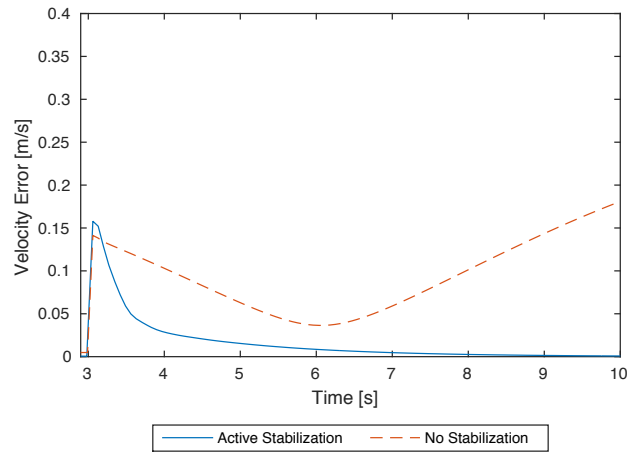


(a) hit with a table (b) hit with a wall

Figure 6: The two simulated situations for the astronaut stabilization task.



(a) hit with a table



(b) hit with a wall

Figure 7: Norm of the linear velocity of Space CoBot, with and without motion stabilization, for the two situations described in the text.

and without the controller. Plots of the translational velocity norm of the robot CoM are shown in Figure 7. It is visible in both cases that the controller successfully stabilizes motion in about 4 seconds. The oscillatory behavior when the controller is disabled corresponds to the velocity, with respect to the inertial frame, of the robot CoM "orbiting" the resultant CoM of astronaut+robot, while it drifts away from the contact point.

All of the simulations discussed in this section were compiled into a video available here:

`https://youtu.be/TQqRAMBv3-M`

7 Conclusions and Future Work

This paper presented both the design and the application of Space CoBot for assistive tasks to astronauts, to operate inside orbiting space stations. It is based on a hexarotor design

with a fully holonomic kinematics. A convergent motion controller was presented, which was then used to explore two applications. The first one is scavenging of free flying debris, while the second one is motion stabilization of astronauts. These application were implemented and evaluated with a realistic robotics simulator.

Future work will proceed in two directions: on the one hand, to build a real prototype to validate approach (partially on Earth, and in full on, e.g., parabolic flight tests), and on the other, to address basic functionalities such as vision-based navigation and human-robot interaction.

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